

1. Setup e Imports

```
import json as _json
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from tqdm import tqdm
import os
import sys
from pathlib import Path

# Agregar path del proyecto para imports
sys.path.append(os.path.abspath('../..'))
from sae.models.sae import SparseAutoencoder
from sae.tools.naming_utils import save_checkpoint, get_model_name, get_extras_id, get_checkpoint_dir, get_model_dir
from sae.tools.experiment_utils import get_experiment_dir, get_plots_dir, get_metrics_dir, get_logs_dir

# Configuración de visualización para Quarto/PDF
pd.set_option('display.width', 100)
pd.set_option('display.max_columns', None)
np.set_printoptions(linewidth=100, precision=4)
os.environ['COLUMNS'] = '100'

# Configuración de reproducibilidad
torch.manual_seed(42)
np.random.seed(42)

# Device
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
```

```
# Print config con formato profesional
config_df = pd.DataFrame({
    'Configuración': ['Device', 'Seed', 'Output Width'],
    'Valor': [str(device), '42', '100']
})

print('\n' + '='*50)
print(' CONFIGURACIÓN INICIAL')
print('='*50)
print(config_df.to_string(index=False))
print('='*50)
```

```
=====
CONFIGURACIÓN INICIAL
=====
Configuración Valor
      Device  cuda
      Seed    42
Output Width  100
=====
```

2 Metadata para Naming de Checkpoints

```
# Metadata para naming de checkpoints (separada de hiperparámetros)
NAMING_META = {
    'variante': 'standard',          # Arquitectura del SAE
    'tecnica': 'l1-decay',           # L1 con penalty annealing
    'capa': 6,                       # Layer del modelo OthelloGPT
    'juegos': 1000,                  # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt', # Path base para cl
    'experiment_base_path': os.path.abspath('../../experiments') ,      # sae/experiments,
    'extras': {
        'expansion': 32,
        'sparsity_coef': 2e-3,
        'epochs': 200,
        'patience': 20,
        'learning_rate': 3e-4,
        'sparsity_decay': 0.98,
```

```

        'batch_size': 256
    }
}

print('\n Metadata de naming:')
for key, value in NAMING_META.items():
    print(f' {key}: {value}')

```

```

Metadata de naming:
variante: standard
tecnica: l1-decay
capa: 6
juegos: 1000
base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
extras: {'expansion': 32, 'sparsity_coef': 0.002, 'epochs': 200, 'patience': 20, 'learning

```

```

extras_id = get_extras_id(
    get_checkpoint_dir(
        NAMING_META['base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos']
    ),
    NAMING_META['extras']
) if NAMING_META.get('extras') else None

print(f'extras_id: {extras_id}')

```

```

extras_id: ex3

```

3. Configuración del SAE

Las activaciones ya fueron extraídas previamente con `sae/activations/run_extraction.py`. Aquí especificamos qué archivo cargar y los hiperparámetros del SAE.

```

# Configuración del SAE
CONFIG = {
    # Arquitectura
    'input_dim': 512,
    'hidden_dim': 16384,
    'tied_weights': False,

    # Entrenamiento
    'batch_size': 256,
    'num_epochs': 200,
    'learning_rate': 3e-4,
    'sparsity_coef': 2e-3,
    'sparsity_decay': 0.98,

    # Early Stopping
    'early_stopping': True,
    'patience': 20,

    # Datos
    'activations_file': Path(f"../../activations/data/layer{NAMING_META['capa']}-{NAMING_META['juegos']}.npy"),
    'save_dir': Path('./saved_model'),
    'plots_dir': get_plots_dir(
        NAMING_META['experiment_base_path'],
        NAMING_META['variante'],
        NAMING_META['tecnica'],
        NAMING_META['capa'],
        NAMING_META['juegos'],
        extras_id=extras_id
    )
}

# Crear directorio local del modelo (plots_dir ya fue creado por get_plots_dir)
CONFIG['save_dir'].mkdir(parents=True, exist_ok=True)

# Mostrar configuración con formato profesional
config_data = {
    'Categoría': [],
    'Parámetro': [],
    'Valor': []
}

```

```

# Arquitectura
for key in ['input_dim', 'hidden_dim', 'tied_weights']:
    config_data['Categoría'].append('Arquitectura')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

# Entrenamiento
for key in ['batch_size', 'num_epochs', 'learning_rate', 'sparsity_coef', 'sparsity_decay']:
    config_data['Categoría'].append('Entrenamiento')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

# Early Stopping
for key in ['early_stopping', 'patience']:
    config_data['Categoría'].append('Early Stopping')
    config_data['Parámetro'].append(key)
    config_data['Valor'].append(str(CONFIG[key]))

config_table = pd.DataFrame(config_data)

print('\n' + '='*70)
print(' CONFIGURACIÓN DEL SAE')
print('='*70)
print(config_table.to_string(index=False))
print('='*70)

print(f'\nArchivo de activaciones: {CONFIG["activations_file"]}')
print(f'Expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f'Plots dir: {CONFIG["plots_dir"]}')
print('='*70)

```

```

=====
CONFIGURACIÓN DEL SAE
=====

```

Categoría	Parámetro	Valor
Arquitectura	input_dim	512
Arquitectura	hidden_dim	16384
Arquitectura	tied_weights	False
Entrenamiento	batch_size	256
Entrenamiento	num_epochs	200
Entrenamiento	learning_rate	0.0003

```

Entrenamiento sparsity_coef 0.002
Entrenamiento sparsity_decay 0.98
Early Stopping early_stopping True
Early Stopping patience 20
=====

Archivo de activaciones: ..\..\..\activations\data\layer6_1000games.npy
Expansión: 32.0x
Plots dir: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\1000games\ex3\plots
=====

```

4. Dataset de Activaciones (Train/Validation Split 80-20%)

```

class ActivationsDataset(Dataset):
    """Dataset para cargar activaciones extraídas"""

    def __init__(self, activations_path):
        self.activations = np.load(activations_path)

    def __len__(self):
        return len(self.activations)

    def __getitem__(self, idx):
        return torch.tensor(self.activations[idx], dtype=torch.float32)

# Cargar dataset completo
full_dataset = ActivationsDataset(CONFIG['activations_file'])

# Información del dataset con formato profesional
dataset_info = pd.DataFrame({
    'Propiedad': ['Total muestras', 'Shape', 'Dtype', 'Tamaño en memoria'],
    'Valor': [
        f"{len(full_dataset):,}",
        str(full_dataset.activations.shape),
        str(full_dataset.activations.dtype),
        f"{full_dataset.activations.nbytes / 1024**2:.2f} MB"
    ]
})

print('\n' + '='*50)
print(' DATASET DE ACTIVACIONES')

```

```

print('='*50)
print(dataset_info.to_string(index=False))
print('='*50)

# Split train/validation (80-20)
train_size = int(0.8 * len(full_dataset))
val_size = len(full_dataset) - train_size
train_dataset, val_dataset = torch.utils.data.random_split(
    full_dataset,
    [train_size, val_size],
    generator=torch.Generator().manual_seed(42)
)

# Resumen del split con formato profesional
split_info = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):,}",
        f"{len(val_dataset):,}",
        f"{len(full_dataset):,}"
    ],
    'Porcentaje': [
        f"{len(train_dataset)/len(full_dataset):.1%}",
        f"{len(val_dataset)/len(full_dataset):.1%}",
        "100.0%"
    ]
})
print('\n' + '='*50)
print(' SPLIT DE DATOS (80-20)')
print('='*50)
print(split_info.to_string(index=False))
print('='*50)

# Crear DataLoaders
train_loader = DataLoader(
    train_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=True,
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)

```

```

val_loader = DataLoader(
    val_dataset,
    batch_size=CONFIG['batch_size'],
    shuffle=False, # No shuffle para validación
    num_workers=0,
    pin_memory=(device.type == 'cuda')
)

# Resumen de DataLoaders
loader_info = pd.DataFrame({
    'DataLoader': ['Train', 'Validation'],
    'Batches': [len(train_loader), len(val_loader)],
    'Batch Size': [CONFIG['batch_size'], CONFIG['batch_size']],
    'Shuffle': ['Sí', 'No']
})

print('\n' + '='*50)
print(' DATALOADERS')
print('='*50)
print(loader_info.to_string(index=False))
print('='*50)

```

```

=====
DATASET DE ACTIVACIONES
=====

```

	Propiedad	Valor
	Total muestras	59,000
	Shape	(59000, 512)
	Dtype	float32
Tamaño en memoria		115.23 MB

```

=====

=====
SPLIT DE DATOS (80-20)
=====

```

	Conjunto	Muestras	Porcentaje
	Train	47,200	80.0%
Validation		11,800	20.0%
	Total	59,000	100.0%

```

=====

```


=====				
DATALOADERS				
=====				
Loader	Batches	Batch Size	Shuffle	
Train	185	256	Sí	
Validation	47	256	No	
=====				

5. Modelo: Sparse Autoencoder

Modelo importado desde `sae.models.sae.SparseAutoencoder`

Arquitectura:

- **Encoder:** Linear $\rightarrow$ ReLU (sparsity natural)- **Decoder:** Linear (reconstrucción)

```
# Crear modelo desde el módulo
model = SparseAutoencoder(
    input_dim=CONFIG['input_dim'],
    hidden_dim=CONFIG['hidden_dim'],
    tied_weights=CONFIG['tied_weights']
).to(device)

# Calcular parámetros
num_params = sum(p.numel() for p in model.parameters())
encoder_params = sum(p.numel() for p in model.encoder.parameters())
if CONFIG['tied_weights']:
    decoder_params = CONFIG['input_dim'] # Solo bias
else:
    decoder_params = sum(p.numel() for p in model.decoder.parameters())

# Resumen del modelo con formato profesional
model_info = pd.DataFrame({
    'Componente': ['Input', 'Encoder', 'Hidden (Latent)', 'Decoder', 'Output', 'Total'],
    'Dimensión': [
        CONFIG['input_dim'],
        f"{CONFIG['input_dim']} → {CONFIG['hidden_dim']}",
        CONFIG['hidden_dim'],
        f"{CONFIG['hidden_dim']} → {CONFIG['input_dim']}",
        CONFIG['input_dim'],
        '_'
    ],
    'Parámetros': [
```

```

        '-',
        f"{encoder_params:,}",
        '-',
        f"{decoder_params:,}",
        '-',
        f"{num_params:,}"
    ]
})

print('\n' + '='*70)
print(' ARQUITECTURA DEL MODELO')
print('='*70)
print(model_info.to_string(index=False))
print('='*70)
print(f'\n Factor de expansión: {CONFIG["hidden_dim"]/CONFIG["input_dim"]:.1f}x')
print(f' Tied weights: {"Sí" if CONFIG["tied_weights"] else "No"}')
print(f' Device: {device}')
print('='*70)

```

```

=====
ARQUITECTURA DEL MODELO
=====

```

	Componente	Dimensión	Parámetros
	Input	512	-
	Encoder	512 → 16384	8,404,992
Hidden	(Latent)	16384	-
	Decoder	16384 → 512	8,389,120
	Output	512	-
	Total	-	16,794,112

```

=====

Factor de expansión: 32.0x
Tied weights: No
Device: cuda
=====

```

6. Función de Pérdida

Loss = MSE(reconstrucción) + λ × L1(activaciones)

- **MSE:** Mide qué tan bien reconstruimos las activaciones originales
- **L1:** Penaliza activaciones grandes → fuerza sparsity

```
def loss_function(x, reconstruction, hidden, sparsity_coef):
    """
    Pérdida compuesta: MSE + L1 penalty

    Args:
        x: Activaciones originales
        reconstruction: Activaciones reconstruidas
        hidden: Activaciones latentes (sparse)
        sparsity_coef: Peso de la penalización L1
    """
    # Error de reconstrucción
    mse_loss = F.mse_loss(reconstruction, x)

    # Sumar sobre las features (dim=1), promediar sobre el batch
    sparsity_loss = hidden.abs().sum(dim=1).mean()

    # Pérdida total
    total_loss = mse_loss + sparsity_coef * sparsity_loss

    return total_loss, mse_loss, sparsity_loss

# Optimizador
optimizer = torch.optim.Adam(model.parameters(), lr=CONFIG['learning_rate'])

# Resumen del optimizador
optim_info = pd.DataFrame({
    'Parámetro': ['Optimizador', 'Learning Rate', 'Sparsity Coef (inicial)', 'Sparsity Decay'],
    'Valor': [
        'Adam',
        f"{CONFIG['learning_rate']:.0e}",
        f"{CONFIG['sparsity_coef']:.0e}",
        f"{CONFIG['sparsity_decay']:.2f}"
    ]
})

print('\n' + '='*50)
print(' OPTIMIZACIÓN')
print('='*50)
print(optim_info.to_string(index=False))
print('='*50)
```

```
=====
OPTIMIZACIÓN
=====
                Parámetro Valor
            Optimizador  Adam
            Learning Rate 3e-04
Sparsity Coef (inicial) 2e-03
            Sparsity Decay 0.98
=====
```

7. Loop de Entrenamiento

```
# Historial de entrenamiento
history = {
    'train_total_loss': [],
    'train_mse_loss': [],
    'train_sparsity_loss': [],
    'train_l0_sparsity': [],
    'train_fraction_active': [],
    'val_total_loss': [],
    'val_mse_loss': [],
    'val_sparsity_loss': [],
    'val_l0_sparsity': [],
    'val_fraction_active': [],
    'sparsity_coef_history': [] # Guardar evolución del coeficiente L1
}

# Early stopping variables
best_val_mse = float('inf')
best_epoch = 0
epochs_no_improve = 0

print(f'\n Iniciando entrenamiento: {CONFIG["num_epochs"]} épocas máximo')
print(f' L1 decay: {CONFIG["sparsity_coef"]:.2e} × {CONFIG["sparsity_decay"]:.2f}^época')
print(f' Early stopping: patience={CONFIG["patience"]} épocas')
print('=' * 70)

for epoch in range(CONFIG['num_epochs']):
```

```

# ===== L1 DECAY =====
# Reducir sparsity_coef exponencialmente
current_sparsity_coef = CONFIG['sparsity_coef'] * (CONFIG['sparsity_decay'] ** epoch)
history['sparsity_coef_history'].append(current_sparsity_coef)

# ===== ENTRENAMIENTO =====
model.train()

epoch_total_loss = 0
epoch_mse_loss = 0
epoch_sparsity_loss = 0
epoch_l0 = 0
epoch_fraction_active = 0

# Barra de progreso
pbar = tqdm(train_loader, desc=f'Época {epoch+1}/{CONFIG["num_epochs"]} [Train] L1={current_sparsity_coef}')

for batch in pbar:
    # Mover al device
    x = batch.to(device)

    # Forward pass
    reconstruction, hidden = model(x)

    # Calcular pérdida con sparsity_coef actualizado
    total_loss, mse_loss, sparsity_loss = loss_function(
        x, reconstruction, hidden, current_sparsity_coef
    )

    # Backward pass
    optimizer.zero_grad()
    total_loss.backward()

    optimizer.step()

    # Métricas
    with torch.no_grad():
        l0 = (hidden > 0).float().sum(dim=1).mean().item()
        fraction_active = (hidden > 0).float().mean().item()

    # Acumular
    epoch_total_loss += total_loss.item()

```

```

epoch_mse_loss += mse_loss.item()
epoch_sparsity_loss += sparsity_loss.item()
epoch_l0 += l0
epoch_fraction_active += fraction_active

# Actualizar barra
pbar.set_postfix({
    'loss': f'{total_loss.item():.4f}',
    'mse': f'{mse_loss.item():.4f}',
    'l0': f'{l0:.1f}'
})

# Promedios de entrenamiento
num_train_batches = len(train_loader)
avg_train_total_loss = epoch_total_loss / num_train_batches
avg_train_mse_loss = epoch_mse_loss / num_train_batches
avg_train_sparsity_loss = epoch_sparsity_loss / num_train_batches
avg_train_l0 = epoch_l0 / num_train_batches
avg_train_fraction_active = epoch_fraction_active / num_train_batches

# ===== VALIDACIÓN =====
model.eval()

val_total_loss = 0
val_mse_loss = 0
val_sparsity_loss = 0
val_l0 = 0
val_fraction_active = 0

with torch.no_grad():
    for batch in val_loader:
        x = batch.to(device)

        # Forward pass
        reconstruction, hidden = model(x)

        # Calcular pérdida con sparsity_coef actualizado
        total_loss, mse_loss, sparsity_loss = loss_function(
            x, reconstruction, hidden, current_sparsity_coef
        )

# Métricas

```

```

10 = (hidden > 0).float().sum(dim=1).mean().item()
fraction_active = (hidden > 0).float().mean().item()

# Acumular
val_total_loss += total_loss.item()
val_mse_loss += mse_loss.item()
val_sparsity_loss += sparsity_loss.item()
val_l0 += l0
val_fraction_active += fraction_active

# Promedios de validación
num_val_batches = len(val_loader)
avg_val_total_loss = val_total_loss / num_val_batches
avg_val_mse_loss = val_mse_loss / num_val_batches
avg_val_sparsity_loss = val_sparsity_loss / num_val_batches
avg_val_l0 = val_l0 / num_val_batches
avg_val_fraction_active = val_fraction_active / num_val_batches

# Guardar en historial
history['train_total_loss'].append(avg_train_total_loss)
history['train_mse_loss'].append(avg_train_mse_loss)
history['train_sparsity_loss'].append(avg_train_sparsity_loss)
history['train_l0_sparsity'].append(avg_train_l0)
history['train_fraction_active'].append(avg_train_fraction_active)

history['val_total_loss'].append(avg_val_total_loss)
history['val_mse_loss'].append(avg_val_mse_loss)
history['val_sparsity_loss'].append(avg_val_sparsity_loss)
history['val_l0_sparsity'].append(avg_val_l0)
history['val_fraction_active'].append(avg_val_fraction_active)

# ===== EARLY STOPPING =====
if CONFIG['early_stopping']:
    if avg_val_mse_loss < best_val_mse:
        best_val_mse = avg_val_mse_loss
        best_epoch = epoch + 1
        epochs_no_improve = 0

    # Guardar mejor modelo (local + organizado)
    best_checkpoint = {
        'model_state_dict': model.state_dict(),
        'optimizer_state_dict': optimizer.state_dict(),

```

```

        'config': CONFIG,
        'history': history,
        'epoch': best_epoch,
        'val_mse': best_val_mse
    }

    # Guardado local
    best_model_name = get_model_name(
        variante=NAMING_META['variante'],
        tecnica=NAMING_META['tecnica'],
        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best'
    )
    torch.save(best_checkpoint, CONFIG['save_dir'] / best_model_name)

    # Guardado organizado
    save_checkpoint(
        model,
        base_path=NAMING_META['base_path'],
        variante=NAMING_META['variante'],
        tecnica=NAMING_META['tecnica'],
        capa=NAMING_META['capa'],
        juegos=NAMING_META['juegos'],
        estado='best',
        extras=NAMING_META['extras'],
        config=CONFIG,
        history=history,
        optimizer=optimizer
    )

    print(f' Mejor modelo guardado en época {best_epoch} (Val MSE: {best_val_mse:.4f})')
else:
    epochs_no_improve += 1
    if epochs_no_improve >= CONFIG['patience']:
        print(f'\n Early stopping activado en época {epoch+1}')
        print(f' Mejor modelo: época {best_epoch} con Val MSE = {best_val_mse:.4f}')
        break

# Log cada 5 épocas
if (epoch + 1) % 5 == 0 or epoch == 0:
    print(f'\n Época {epoch+1}/{CONFIG["num_epochs"]} | L1 coef: {current_sparsity_coef:}')

```



```

        print(f'  Train - Total: {avg_train_total_loss:.4f} | MSE: {avg_train_mse_loss:.4f}')
        print(f'  Val   - Total: {avg_val_total_loss:.4f} | MSE: {avg_val_mse_loss:.4f} | L0')
        if CONFIG['early_stopping']:
            print(f'    Best Val MSE: {best_val_mse:.4f} @ época {best_epoch} | Sin mejora: {best_val_mse:.4f}')

print('\n' + '='*70)
print(' RESUMEN DEL ENTRENAMIENTO')
print('='*70)

if CONFIG['early_stopping']:
    print(f'Mejor modelo en época {best_epoch} con Val MSE = {best_val_mse:.4f}')
    print(f'Early stopping tras {len(history["train_total_loss"])} épocas')
else:
    print(f'Entrenamiento completo: {CONFIG["num_epochs"]} épocas')

# Tabla de métricas finales
final_metrics = pd.DataFrame({
    'Métrica': ['Total Loss', 'MSE Loss', 'L0 Sparsity', 'Fraction Active'],
    'Train': [
        f"{history['train_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['train_fraction_active'][-1]:.2%}"
    ],
    'Validation': [
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['val_l0_sparsity'][-1]:.1f}",
        f"{history['val_fraction_active'][-1]:.2%}"
    ]
})

print('\nMétricas Finales (Última Época):')
print(final_metrics.to_string(index=False))
print('='*70)

```

Iniciando entrenamiento: 200 épocas máximo
 L1 decay: $2.00e-03 \times 0.98^{\text{época}}$
 Early stopping: patience=20 épocas

=====

Época 1/200 [Train] L1=2.00e-03: 100%| | 185/185 [00:02<00:00, 70.77it/s, loss=0.5980, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 1 (Val MSE: 0.5321)

Época 1/200 | L1 coef: 2.00e-03

Train - Total: 0.7915 | MSE: 0.6086 | L0: 619.0

Val - Total: 0.5916 | MSE: 0.5321 | L0: 103.4

Best Val MSE: 0.5321 @ época 1 | Sin mejora: 0/20

Época 2/200 [Train] L1=1.96e-03: 100%| | 185/185 [00:02<00:00, 80.51it/s, loss=0.5011, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 2 (Val MSE: 0.4074)

Época 3/200 [Train] L1=1.92e-03: 100%| | 185/185 [00:02<00:00, 78.21it/s, loss=0.4099, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 3 (Val MSE: 0.3147)

Época 4/200 [Train] L1=1.88e-03: 100%| | 185/185 [00:02<00:00, 79.34it/s, loss=0.3742, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 4 (Val MSE: 0.2661)

Época 5/200 [Train] L1=1.84e-03: 100%| | 185/185 [00:02<00:00, 80.14it/s, loss=0.3440, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 5 (Val MSE: 0.2323)

Época 5/200 | L1 coef: 1.84e-03

Train - Total: 0.3482 | MSE: 0.2443 | L0: 133.2

Val - Total: 0.3391 | MSE: 0.2323 | L0: 152.7

Best Val MSE: 0.2323 @ época 5 | Sin mejora: 0/20

Época 6/200 [Train] L1=1.81e-03: 100%| | 185/185 [00:01<00:00, 95.49it/s, loss=0.3049, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 6 (Val MSE: 0.2043)

Época 7/200 [Train] L1=1.77e-03: 100%| | 185/185 [00:01<00:00, 93.69it/s, loss=0.2918, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 7 (Val MSE: 0.1816)

Época 8/200 [Train] L1=1.74e-03: 100%| | 185/185 [00:01<00:00, 92.57it/s, loss=0.2668, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 8 (Val MSE: 0.1620)

Época 9/200 [Train] L1=1.70e-03: 100%| | 185/185 [00:02<00:00, 89.68it/s, loss=0.2562, m

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 9 (Val MSE: 0.1448)

Época 10/200 [Train] L1=1.67e-03: 100%| | 185/185 [00:01<00:00, 92.66it/s, loss=0.2436,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 10 (Val MSE: 0.1299)

Época 10/200 | L1 coef: 1.67e-03

Train - Total: 0.2430 | MSE: 0.1317 | L0: 235.4

Val - Total: 0.2428 | MSE: 0.1299 | L0: 255.0

Best Val MSE: 0.1299 @ época 10 | Sin mejora: 0/20

Época 11/200 [Train] L1=1.63e-03: 100%| | 185/185 [00:02<00:00, 92.33it/s, loss=0.2215,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 11 (Val MSE: 0.1167)

Época 12/200 [Train] L1=1.60e-03: 100%| | 185/185 [00:02<00:00, 86.35it/s, loss=0.2105,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 12 (Val MSE: 0.1045)

Época 13/200 [Train] L1=1.57e-03: 100%| | 185/185 [00:02<00:00, 84.21it/s, loss=0.1987,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 13 (Val MSE: 0.0939)

Época 14/200 [Train] L1=1.54e-03: 100%| | 185/185 [00:02<00:00, 78.04it/s, loss=0.1908,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 14 (Val MSE: 0.0850)

Época 15/200 [Train] L1=1.51e-03: 100%| | 185/185 [00:02<00:00, 81.87it/s, loss=0.1795,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 15 (Val MSE: 0.0774)

Época 15/200 | L1 coef: 1.51e-03

Train - Total: 0.1817 | MSE: 0.0756 | L0: 321.2

Val - Total: 0.1844 | MSE: 0.0774 | L0: 338.6

Best Val MSE: 0.0774 @ época 15 | Sin mejora: 0/20

Época 16/200 [Train] L1=1.48e-03: 100%| | 185/185 [00:02<00:00, 80.72it/s, loss=0.1666,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 16 (Val MSE: 0.0710)

Época 17/200 [Train] L1=1.45e-03: 100%| | 185/185 [00:02<00:00, 78.91it/s, loss=0.1683,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 17 (Val MSE: 0.0652)

Época 18/200 [Train] L1=1.42e-03: 100%| | 185/185 [00:02<00:00, 87.56it/s, loss=0.1542,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 18 (Val MSE: 0.0609)

Época 19/200 [Train] L1=1.39e-03: 100%| | 185/185 [00:02<00:00, 80.61it/s, loss=0.1501,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 19 (Val MSE: 0.0576)

Época 20/200 [Train] L1=1.36e-03: 100%| | 185/185 [00:02<00:00, 78.16it/s, loss=0.1397,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 20 (Val MSE: 0.0540)

Época 20/200 | L1 coef: 1.36e-03

Train - Total: 0.1435 | MSE: 0.0500 | L0: 373.1

Val - Total: 0.1478 | MSE: 0.0540 | L0: 386.5

Best Val MSE: 0.0540 @ época 20 | Sin mejora: 0/20

Época 21/200 [Train] L1=1.34e-03: 100%| | 185/185 [00:02<00:00, 82.50it/s, loss=0.1347,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 21 (Val MSE: 0.0508)

Época 22/200 [Train] L1=1.31e-03: 100%| | 185/185 [00:02<00:00, 78.68it/s, loss=0.1284,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 22 (Val MSE: 0.0482)

Época 23/200 [Train] L1=1.28e-03: 100%| | 185/185 [00:02<00:00, 77.70it/s, loss=0.1253,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 23 (Val MSE: 0.0455)

Época 24/200 [Train] L1=1.26e-03: 100%| | 185/185 [00:02<00:00, 76.57it/s, loss=0.1186,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 24 (Val MSE: 0.0438)

Época 25/200 [Train] L1=1.23e-03: 100%| | 185/185 [00:02<00:00, 78.38it/s, loss=0.1168,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 25 (Val MSE: 0.0415)

Época 25/200 | L1 coef: 1.23e-03

Train - Total: 0.1178 | MSE: 0.0371 | L0: 404.1

Val - Total: 0.1226 | MSE: 0.0415 | L0: 416.7

Best Val MSE: 0.0415 @ época 25 | Sin mejora: 0/20

Época 26/200 [Train] L1=1.21e-03: 100%| | 185/185 [00:02<00:00, 79.96it/s, loss=0.1134,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 26 (Val MSE: 0.0399)

Época 27/200 [Train] L1=1.18e-03: 100%| | 185/185 [00:02<00:00, 79.19it/s, loss=0.1085,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 27 (Val MSE: 0.0377)

Época 28/200 [Train] L1=1.16e-03: 100%| | 185/185 [00:02<00:00, 90.18it/s, loss=0.1045,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 28 (Val MSE: 0.0366)

Época 29/200 [Train] L1=1.14e-03: 100%| | 185/185 [00:02<00:00, 91.48it/s, loss=0.1002,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 29 (Val MSE: 0.0349)

Época 30/200 [Train] L1=1.11e-03: 100%| | 185/185 [00:02<00:00, 91.63it/s, loss=0.0989,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 30 (Val MSE: 0.0335)

Época 30/200 | L1 coef: 1.11e-03

Train - Total: 0.0984 | MSE: 0.0288 | L0: 425.8

Val - Total: 0.1035 | MSE: 0.0335 | L0: 437.3

Best Val MSE: 0.0335 @ época 30 | Sin mejora: 0/20

Época 31/200 [Train] L1=1.09e-03: 100%| | 185/185 [00:02<00:00, 90.58it/s, loss=0.0950,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 31 (Val MSE: 0.0321)

Época 32/200 [Train] L1=1.07e-03: 100%| | 185/185 [00:02<00:00, 87.41it/s, loss=0.0918,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 32 (Val MSE: 0.0313)

Época 33/200 [Train] L1=1.05e-03: 100%| | 185/185 [00:02<00:00, 91.18it/s, loss=0.0906,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 33 (Val MSE: 0.0296)

Época 34/200 [Train] L1=1.03e-03: 100%| | 185/185 [00:02<00:00, 80.25it/s, loss=0.0877,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 34 (Val MSE: 0.0289)

Época 35/200 [Train] L1=1.01e-03: 100%| | 185/185 [00:02<00:00, 78.98it/s, loss=0.0816,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 35 (Val MSE: 0.0279)

Época 35/200 | L1 coef: 1.01e-03
Train - Total: 0.0834 | MSE: 0.0230 | L0: 442.5
Val - Total: 0.0884 | MSE: 0.0279 | L0: 452.6
Best Val MSE: 0.0279 @ época 35 | Sin mejora: 0/20

Época 36/200 [Train] L1=9.86e-04: 100%| | 185/185 [00:02<00:00, 80.18it/s, loss=0.0805,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 36 (Val MSE: 0.0267)

Época 37/200 [Train] L1=9.66e-04: 100%| | 185/185 [00:02<00:00, 79.41it/s, loss=0.0788,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 37 (Val MSE: 0.0254)

Época 38/200 [Train] L1=9.47e-04: 100%| | 185/185 [00:02<00:00, 77.52it/s, loss=0.0773,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 38 (Val MSE: 0.0252)

Época 39/200 [Train] L1=9.28e-04: 100%| | 185/185 [00:02<00:00, 84.26it/s, loss=0.0750,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 39 (Val MSE: 0.0241)

Época 40/200 [Train] L1=9.10e-04: 100%| | 185/185 [00:02<00:00, 80.15it/s, loss=0.0708,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 40 (Val MSE: 0.0230)

Época 40/200 | L1 coef: 9.10e-04
Train - Total: 0.0715 | MSE: 0.0188 | L0: 456.1
Val - Total: 0.0761 | MSE: 0.0230 | L0: 467.2
Best Val MSE: 0.0230 @ época 40 | Sin mejora: 0/20

Época 41/200 [Train] L1=8.91e-04: 100%| | 185/185 [00:02<00:00, 82.92it/s, loss=0.0704,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 41 (Val MSE: 0.0224)

Época 42/200 [Train] L1=8.74e-04: 100%| | 185/185 [00:02<00:00, 80.38it/s, loss=0.0672,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 42 (Val MSE: 0.0217)

Época 43/200 [Train] L1=8.56e-04: 100%| | 185/185 [00:02<00:00, 82.18it/s, loss=0.0669,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 43 (Val MSE: 0.0211)

Época 44/200 [Train] L1=8.39e-04: 100%| | 185/185 [00:02<00:00, 81.79it/s, loss=0.0641,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 44 (Val MSE: 0.0202)

Época 45/200 [Train] L1=8.22e-04: 100%| | 185/185 [00:02<00:00, 82.46it/s, loss=0.0616,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 45 (Val MSE: 0.0196)

Época 45/200 | L1 coef: 8.22e-04
Train - Total: 0.0620 | MSE: 0.0158 | L0: 466.9
Val - Total: 0.0662 | MSE: 0.0196 | L0: 476.3
Best Val MSE: 0.0196 @ época 45 | Sin mejora: 0/20

Época 46/200 [Train] L1=8.06e-04: 100%| | 185/185 [00:02<00:00, 82.00it/s, loss=0.0603,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 46 (Val MSE: 0.0193)

Época 47/200 [Train] L1=7.90e-04: 100%| | 185/185 [00:02<00:00, 80.47it/s, loss=0.0599,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 47 (Val MSE: 0.0184)

Época 48/200 [Train] L1=7.74e-04: 100%| | 185/185 [00:02<00:00, 80.31it/s, loss=0.0572,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 48 (Val MSE: 0.0179)

Época 49/200 [Train] L1=7.58e-04: 100%| | 185/185 [00:02<00:00, 77.81it/s, loss=0.0559,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 49 (Val MSE: 0.0173)

Época 50/200 [Train] L1=7.43e-04: 100%| | 185/185 [00:02<00:00, 82.16it/s, loss=0.0527,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 50 (Val MSE: 0.0171)

Época 50/200 | L1 coef: 7.43e-04
Train - Total: 0.0540 | MSE: 0.0132 | L0: 475.9
Val - Total: 0.0582 | MSE: 0.0171 | L0: 484.9
Best Val MSE: 0.0171 @ época 50 | Sin mejora: 0/20

Época 51/200 [Train] L1=7.28e-04: 100%| | 185/185 [00:02<00:00, 78.44it/s, loss=0.0537,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 51 (Val MSE: 0.0163)

Época 52/200 [Train] L1=7.14e-04: 100%| | 185/185 [00:02<00:00, 79.83it/s, loss=0.0507,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 52 (Val MSE: 0.0159)

Época 53/200 [Train] L1=6.99e-04: 100%| | 185/185 [00:02<00:00, 91.37it/s, loss=0.0497,
Época 54/200 [Train] L1=6.86e-04: 100%| | 185/185 [00:02<00:00, 87.86it/s, loss=0.0489,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 54 (Val MSE: 0.0151)

Época 55/200 [Train] L1=6.72e-04: 100%| | 185/185 [00:02<00:00, 79.54it/s, loss=0.0456,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 55 (Val MSE: 0.0146)

Época 55/200 | L1 coef: 6.72e-04
Train - Total: 0.0474 | MSE: 0.0112 | L0: 483.4
Val - Total: 0.0511 | MSE: 0.0146 | L0: 492.2
Best Val MSE: 0.0146 @ época 55 | Sin mejora: 0/20

Época 56/200 [Train] L1=6.58e-04: 100%| | 185/185 [00:02<00:00, 81.78it/s, loss=0.0465,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 56 (Val MSE: 0.0142)

Época 57/200 [Train] L1=6.45e-04: 100% | 185/185 [00:02<00:00, 81.48it/s, loss=0.0440,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 57 (Val MSE: 0.0137)

Época 58/200 [Train] L1=6.32e-04: 100% | 185/185 [00:02<00:00, 79.96it/s, loss=0.0452,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 58 (Val MSE: 0.0136)

Época 59/200 [Train] L1=6.20e-04: 100% | 185/185 [00:02<00:00, 83.03it/s, loss=0.0423,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 59 (Val MSE: 0.0133)

Época 60/200 [Train] L1=6.07e-04: 100% | 185/185 [00:02<00:00, 78.45it/s, loss=0.0428,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 60 (Val MSE: 0.0127)

Época 60/200 | L1 coef: 6.07e-04
Train - Total: 0.0420 | MSE: 0.0098 | L0: 490.0
Val - Total: 0.0452 | MSE: 0.0127 | L0: 498.4
Best Val MSE: 0.0127 @ época 60 | Sin mejora: 0/20

Época 61/200 [Train] L1=5.95e-04: 100% | 185/185 [00:02<00:00, 81.31it/s, loss=0.0408,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 61 (Val MSE: 0.0122)

Época 62/200 [Train] L1=5.83e-04: 100%| | 185/185 [00:02<00:00, 81.85it/s, loss=0.0400,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 62 (Val MSE: 0.0120)

Época 63/200 [Train] L1=5.72e-04: 100%| | 185/185 [00:02<00:00, 80.58it/s, loss=0.0383,

Época 64/200 [Train] L1=5.60e-04: 100%| | 185/185 [00:02<00:00, 80.91it/s, loss=0.0387,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 64 (Val MSE: 0.0115)

Época 65/200 [Train] L1=5.49e-04: 100%| | 185/185 [00:02<00:00, 79.54it/s, loss=0.0367,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 65 (Val MSE: 0.0113)

Época 65/200 | L1 coef: 5.49e-04

Train - Total: 0.0373 | MSE: 0.0086 | L0: 495.6

Val - Total: 0.0401 | MSE: 0.0113 | L0: 502.7

Best Val MSE: 0.0113 @ época 65 | Sin mejora: 0/20

Época 66/200 [Train] L1=5.38e-04: 100%| | 185/185 [00:02<00:00, 79.57it/s, loss=0.0371,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 66 (Val MSE: 0.0110)

Época 67/200 [Train] L1=5.27e-04: 100%| | 185/185 [00:02<00:00, 78.22it/s, loss=0.0338,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 67 (Val MSE: 0.0108)

Época 68/200 [Train] L1=5.17e-04: 100%| | 185/185 [00:02<00:00, 81.72it/s, loss=0.0363,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 68 (Val MSE: 0.0104)

Época 69/200 [Train] L1=5.06e-04: 100% | 185/185 [00:02<00:00, 80.67it/s, loss=0.0332,
Época 70/200 [Train] L1=4.96e-04: 100% | 185/185 [00:02<00:00, 82.05it/s, loss=0.0339,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 70 (Val MSE: 0.0100)

Época 70/200 | L1 coef: 4.96e-04
Train - Total: 0.0332 | MSE: 0.0075 | L0: 500.6
Val - Total: 0.0359 | MSE: 0.0100 | L0: 507.5
Best Val MSE: 0.0100 @ época 70 | Sin mejora: 0/20

Época 71/200 [Train] L1=4.86e-04: 100% | 185/185 [00:02<00:00, 79.66it/s, loss=0.0320,
Época 72/200 [Train] L1=4.77e-04: 100% | 185/185 [00:02<00:00, 81.47it/s, loss=0.0314,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 72 (Val MSE: 0.0093)

Época 73/200 [Train] L1=4.67e-04: 100% | 185/185 [00:02<00:00, 81.66it/s, loss=0.0308,
Época 74/200 [Train] L1=4.58e-04: 100% | 185/185 [00:02<00:00, 80.37it/s, loss=0.0304,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 74 (Val MSE: 0.0090)

Época 75/200 [Train] L1=4.48e-04: 100% | 185/185 [00:02<00:00, 80.35it/s, loss=0.0304,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 75 (Val MSE: 0.0089)

Época 75/200 | L1 coef: 4.48e-04
Train - Total: 0.0297 | MSE: 0.0066 | L0: 505.2
Val - Total: 0.0322 | MSE: 0.0089 | L0: 512.4
Best Val MSE: 0.0089 @ época 75 | Sin mejora: 0/20

Época 76/200 [Train] L1=4.40e-04: 100%| | 185/185 [00:02<00:00, 80.35it/s, loss=0.0294,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 76 (Val MSE: 0.0088)

Época 77/200 [Train] L1=4.31e-04: 100%| | 185/185 [00:02<00:00, 82.09it/s, loss=0.0276,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 77 (Val MSE: 0.0086)

Época 78/200 [Train] L1=4.22e-04: 100%| | 185/185 [00:02<00:00, 87.83it/s, loss=0.0286,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 78 (Val MSE: 0.0083)

Época 79/200 [Train] L1=4.14e-04: 100%| | 185/185 [00:02<00:00, 81.14it/s, loss=0.0273,

Época 80/200 [Train] L1=4.05e-04: 100%| | 185/185 [00:02<00:00, 80.00it/s, loss=0.0272,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 80 (Val MSE: 0.0079)

Época 80/200 | L1 coef: 4.05e-04

Train - Total: 0.0268 | MSE: 0.0060 | L0: 508.8

Val - Total: 0.0289 | MSE: 0.0079 | L0: 516.0

Best Val MSE: 0.0079 @ época 80 | Sin mejora: 0/20

Época 81/200 [Train] L1=3.97e-04: 100%| | 185/185 [00:02<00:00, 82.59it/s, loss=0.0268,

Época 82/200 [Train] L1=3.89e-04: 100%| | 185/185 [00:02<00:00, 81.04it/s, loss=0.0261,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 82 (Val MSE: 0.0077)

Época 83/200 [Train] L1=3.82e-04: 100%| | 185/185 [00:02<00:00, 79.17it/s, loss=0.0253,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 83 (Val MSE: 0.0077)

Época 84/200 [Train] L1=3.74e-04: 100%| | 185/185 [00:02<00:00, 81.56it/s, loss=0.0249,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 84 (Val MSE: 0.0075)

Época 85/200 [Train] L1=3.66e-04: 100%| | 185/185 [00:02<00:00, 82.20it/s, loss=0.0248,

Época 85/200 | L1 coef: 3.66e-04
Train - Total: 0.0244 | MSE: 0.0056 | L0: 511.6
Val - Total: 0.0265 | MSE: 0.0076 | L0: 519.1
Best Val MSE: 0.0075 @ época 84 | Sin mejora: 1/20

Época 86/200 [Train] L1=3.59e-04: 100%| | 185/185 [00:02<00:00, 78.00it/s, loss=0.0251,
Época 87/200 [Train] L1=3.52e-04: 100%| | 185/185 [00:02<00:00, 81.09it/s, loss=0.0241,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 87 (Val MSE: 0.0072)

Época 88/200 [Train] L1=3.45e-04: 100%| | 185/185 [00:02<00:00, 78.46it/s, loss=0.0222,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 88 (Val MSE: 0.0070)

Época 89/200 [Train] L1=3.38e-04: 100%| | 185/185 [00:02<00:00, 79.67it/s, loss=0.0229,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 89 (Val MSE: 0.0066)

Época 90/200 [Train] L1=3.31e-04: 100%| | 185/185 [00:02<00:00, 79.56it/s, loss=0.0223,

Época 90/200 | L1 coef: 3.31e-04
Train - Total: 0.0219 | MSE: 0.0049 | L0: 515.0
Val - Total: 0.0239 | MSE: 0.0068 | L0: 521.1
Best Val MSE: 0.0066 @ época 89 | Sin mejora: 1/20

Época 91/200 [Train] L1=3.25e-04: 100%| | 185/185 [00:02<00:00, 80.52it/s, loss=0.0215,
Época 92/200 [Train] L1=3.18e-04: 100%| | 185/185 [00:02<00:00, 78.06it/s, loss=0.0208,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 92 (Val MSE: 0.0066)

Época 93/200 [Train] L1=3.12e-04: 100%| | 185/185 [00:02<00:00, 81.98it/s, loss=0.0207,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 93 (Val MSE: 0.0063)

Época 94/200 [Train] L1=3.06e-04: 100%| | 185/185 [00:02<00:00, 78.25it/s, loss=0.0207,
Época 95/200 [Train] L1=2.99e-04: 100%| | 185/185 [00:02<00:00, 82.61it/s, loss=0.0206,

Época 95/200 | L1 coef: 2.99e-04
Train - Total: 0.0202 | MSE: 0.0049 | L0: 517.2
Val - Total: 0.0225 | MSE: 0.0070 | L0: 524.3
Best Val MSE: 0.0063 @ época 93 | Sin mejora: 2/20

Época 96/200 [Train] L1=2.93e-04: 100%| | 185/185 [00:02<00:00, 77.43it/s, loss=0.0196,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 96 (Val MSE: 0.0059)

Época 97/200 [Train] L1=2.88e-04: 100%| | 185/185 [00:02<00:00, 78.32it/s, loss=0.0196,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 97 (Val MSE: 0.0058)

Época 98/200 [Train] L1=2.82e-04: 100%| | 185/185 [00:02<00:00, 77.26it/s, loss=0.0184,
Época 99/200 [Train] L1=2.76e-04: 100%| | 185/185 [00:02<00:00, 75.47it/s, loss=0.0191,
Época 100/200 [Train] L1=2.71e-04: 100%| | 185/185 [00:02<00:00, 75.10it/s, loss=0.0179,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 100 (Val MSE: 0.0057)

Época 100/200 | L1 coef: 2.71e-04
Train - Total: 0.0181 | MSE: 0.0042 | L0: 519.9
Val - Total: 0.0198 | MSE: 0.0057 | L0: 527.1
Best Val MSE: 0.0057 @ época 100 | Sin mejora: 0/20

Época 101/200 [Train] L1=2.65e-04: 100%| | 185/185 [00:02<00:00, 79.56it/s, loss=0.0185,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 101 (Val MSE: 0.0056)

Época 102/200 [Train] L1=2.60e-04: 100%| | 185/185 [00:02<00:00, 79.86it/s, loss=0.0175,
Época 103/200 [Train] L1=2.55e-04: 100%| | 185/185 [00:02<00:00, 79.10it/s, loss=0.0170,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 103 (Val MSE: 0.0055)

Época 104/200 [Train] L1=2.50e-04: 100%| | 185/185 [00:02<00:00, 78.27it/s, loss=0.0173,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 104 (Val MSE: 0.0054)

Época 105/200 [Train] L1=2.45e-04: 100%| | 185/185 [00:02<00:00, 80.96it/s, loss=0.0159,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 105 (Val MSE: 0.0050)

Época 105/200 | L1 coef: 2.45e-04
Train - Total: 0.0163 | MSE: 0.0037 | L0: 522.2
Val - Total: 0.0178 | MSE: 0.0050 | L0: 529.3
Best Val MSE: 0.0050 @ época 105 | Sin mejora: 0/20

Época 106/200 [Train] L1=2.40e-04: 100%| | 185/185 [00:02<00:00, 77.57it/s, loss=0.0162,
Época 107/200 [Train] L1=2.35e-04: 100%| | 185/185 [00:02<00:00, 77.64it/s, loss=0.0161,
Época 108/200 [Train] L1=2.30e-04: 100%| | 185/185 [00:02<00:00, 75.92it/s, loss=0.0156,
Época 109/200 [Train] L1=2.26e-04: 100%| | 185/185 [00:02<00:00, 77.26it/s, loss=0.0151,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 109 (Val MSE: 0.0050)

Época 110/200 [Train] L1=2.21e-04: 100%| | 185/185 [00:02<00:00, 78.60it/s, loss=0.0148,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 110 (Val MSE: 0.0048)

Época 110/200 | L1 coef: 2.21e-04
Train - Total: 0.0151 | MSE: 0.0036 | L0: 524.1
Val - Total: 0.0165 | MSE: 0.0048 | L0: 531.6
Best Val MSE: 0.0048 @ época 110 | Sin mejora: 0/20

Época 111/200 [Train] L1=2.17e-04: 100%| | 185/185 [00:02<00:00, 80.71it/s, loss=0.0142,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 111 (Val MSE: 0.0047)

Época 112/200 [Train] L1=2.12e-04: 100%| | 185/185 [00:02<00:00, 80.25it/s, loss=0.0137,
Época 113/200 [Train] L1=2.08e-04: 100%| | 185/185 [00:02<00:00, 80.14it/s, loss=0.0146,
Época 114/200 [Train] L1=2.04e-04: 100%| | 185/185 [00:02<00:00, 77.15it/s, loss=0.0141,
Época 115/200 [Train] L1=2.00e-04: 100%| | 185/185 [00:02<00:00, 78.41it/s, loss=0.0138,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_st
decay_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 115 (Val MSE: 0.0045)

Época 115/200 | L1 coef: 2.00e-04
Train - Total: 0.0139 | MSE: 0.0034 | L0: 525.9
Val - Total: 0.0151 | MSE: 0.0045 | L0: 533.3
Best Val MSE: 0.0045 @ época 115 | Sin mejora: 0/20

Época 116/200 [Train] L1=1.96e-04: 100%| | 185/185 [00:02<00:00, 81.27it/s, loss=0.0135,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 116 (Val MSE: 0.0045)

Época 117/200 [Train] L1=1.92e-04: 100%| | 185/185 [00:02<00:00, 78.92it/s, loss=0.0135,

Época 118/200 [Train] L1=1.88e-04: 100%| | 185/185 [00:02<00:00, 79.65it/s, loss=0.0132,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 118 (Val MSE: 0.0043)

Época 119/200 [Train] L1=1.84e-04: 100%| | 185/185 [00:02<00:00, 77.11it/s, loss=0.0131,

Época 120/200 [Train] L1=1.81e-04: 100%| | 185/185 [00:02<00:00, 80.44it/s, loss=0.0130,

Época 120/200 | L1 coef: 1.81e-04

Train - Total: 0.0128 | MSE: 0.0033 | L0: 527.7

Val - Total: 0.0145 | MSE: 0.0049 | L0: 534.3

Best Val MSE: 0.0043 @ época 118 | Sin mejora: 2/20

Época 121/200 [Train] L1=1.77e-04: 100%| | 185/185 [00:02<00:00, 76.04it/s, loss=0.0124,

Época 122/200 [Train] L1=1.74e-04: 100%| | 185/185 [00:02<00:00, 77.16it/s, loss=0.0125,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 122 (Val MSE: 0.0042)

Época 123/200 [Train] L1=1.70e-04: 100%| | 185/185 [00:02<00:00, 81.69it/s, loss=0.0118,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 123 (Val MSE: 0.0038)

Época 124/200 [Train] L1=1.67e-04: 100%| | 185/185 [00:02<00:00, 79.06it/s, loss=0.0120,

Época 125/200 [Train] L1=1.63e-04: 100%| | 185/185 [00:02<00:00, 78.25it/s, loss=0.0118,

Época 125/200 | L1 coef: 1.63e-04
Train - Total: 0.0115 | MSE: 0.0029 | L0: 529.2
Val - Total: 0.0127 | MSE: 0.0040 | L0: 536.9
Best Val MSE: 0.0038 @ época 123 | Sin mejora: 2/20

Época 126/200 [Train] L1=1.60e-04: 100%| | 185/185 [00:02<00:00, 79.63it/s, loss=0.0119,
Época 127/200 [Train] L1=1.57e-04: 100%| | 185/185 [00:02<00:00, 80.55it/s, loss=0.0114,
Época 128/200 [Train] L1=1.54e-04: 100%| | 185/185 [00:02<00:00, 78.23it/s, loss=0.0111,
Época 129/200 [Train] L1=1.51e-04: 100%| | 185/185 [00:02<00:00, 78.15it/s, loss=0.0106,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 129 (Val MSE: 0.0037)

Época 130/200 [Train] L1=1.48e-04: 100%| | 185/185 [00:02<00:00, 80.42it/s, loss=0.0105,

Época 130/200 | L1 coef: 1.48e-04
Train - Total: 0.0106 | MSE: 0.0027 | L0: 530.5
Val - Total: 0.0116 | MSE: 0.0037 | L0: 537.9
Best Val MSE: 0.0037 @ época 129 | Sin mejora: 1/20

Época 131/200 [Train] L1=1.45e-04: 100%| | 185/185 [00:02<00:00, 82.01it/s, loss=0.0108,
Época 132/200 [Train] L1=1.42e-04: 100%| | 185/185 [00:02<00:00, 80.99it/s, loss=0.0108,
Época 133/200 [Train] L1=1.39e-04: 100%| | 185/185 [00:02<00:00, 82.63it/s, loss=0.0100,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 133 (Val MSE: 0.0035)

Época 134/200 [Train] L1=1.36e-04: 100%| | 185/185 [00:02<00:00, 80.26it/s, loss=0.0104,
Época 135/200 [Train] L1=1.33e-04: 100%| | 185/185 [00:02<00:00, 78.24it/s, loss=0.0102,

Época 135/200 | L1 coef: 1.33e-04
Train - Total: 0.0099 | MSE: 0.0027 | L0: 531.6
Val - Total: 0.0109 | MSE: 0.0037 | L0: 538.5
Best Val MSE: 0.0035 @ época 133 | Sin mejora: 2/20

Época 136/200 [Train] L1=1.31e-04: 100%| | 185/185 [00:02<00:00, 76.48it/s, loss=0.0100,
Época 137/200 [Train] L1=1.28e-04: 100%| | 185/185 [00:02<00:00, 77.72it/s, loss=0.0091,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 137 (Val MSE: 0.0034)

Época 138/200 [Train] L1=1.26e-04: 100%| | 185/185 [00:02<00:00, 76.55it/s, loss=0.0094,
Época 139/200 [Train] L1=1.23e-04: 100%| | 185/185 [00:02<00:00, 79.34it/s, loss=0.0092,
Época 140/200 [Train] L1=1.21e-04: 100%| | 185/185 [00:02<00:00, 75.40it/s, loss=0.0091,

Época 140/200 | L1 coef: 1.21e-04

Train - Total: 0.0091 | MSE: 0.0025 | L0: 532.6

Val - Total: 0.0101 | MSE: 0.0035 | L0: 539.2

Best Val MSE: 0.0034 @ época 137 | Sin mejora: 3/20

Época 141/200 [Train] L1=1.18e-04: 100%| | 185/185 [00:02<00:00, 78.75it/s, loss=0.0089,
Época 142/200 [Train] L1=1.16e-04: 100%| | 185/185 [00:02<00:00, 74.55it/s, loss=0.0087,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 142 (Val MSE: 0.0033)

Época 143/200 [Train] L1=1.14e-04: 100%| | 185/185 [00:02<00:00, 78.62it/s, loss=0.0082,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 143 (Val MSE: 0.0032)

Época 144/200 [Train] L1=1.11e-04: 100%| | 185/185 [00:02<00:00, 79.23it/s, loss=0.0082,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 144 (Val MSE: 0.0031)

Época 145/200 [Train] L1=1.09e-04: 100%| | 185/185 [00:02<00:00, 79.47it/s, loss=0.0085,

Época 145/200 | L1 coef: 1.09e-04
Train - Total: 0.0083 | MSE: 0.0023 | L0: 533.9
Val - Total: 0.0094 | MSE: 0.0034 | L0: 540.9
Best Val MSE: 0.0031 @ época 144 | Sin mejora: 1/20

Época 146/200 [Train] L1=1.07e-04: 100%| | 185/185 [00:02<00:00, 78.28it/s, loss=0.0089,
Época 147/200 [Train] L1=1.05e-04: 100%| | 185/185 [00:02<00:00, 78.16it/s, loss=0.0078,
Época 148/200 [Train] L1=1.03e-04: 100%| | 185/185 [00:02<00:00, 80.04it/s, loss=0.0079,
Época 149/200 [Train] L1=1.01e-04: 100%| | 185/185 [00:02<00:00, 76.96it/s, loss=0.0078,
Época 150/200 [Train] L1=9.86e-05: 100%| | 185/185 [00:02<00:00, 78.09it/s, loss=0.0076,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 150 (Val MSE: 0.0030)

Época 150/200 | L1 coef: 9.86e-05
Train - Total: 0.0078 | MSE: 0.0023 | L0: 534.8
Val - Total: 0.0085 | MSE: 0.0030 | L0: 541.8
Best Val MSE: 0.0030 @ época 150 | Sin mejora: 0/20

Época 151/200 [Train] L1=9.66e-05: 100%| | 185/185 [00:02<00:00, 83.22it/s, loss=0.0077,
Época 152/200 [Train] L1=9.47e-05: 100%| | 185/185 [00:02<00:00, 81.66it/s, loss=0.0074,
Época 153/200 [Train] L1=9.28e-05: 100%| | 185/185 [00:02<00:00, 81.91it/s, loss=0.0073,
Época 154/200 [Train] L1=9.09e-05: 100%| | 185/185 [00:02<00:00, 79.38it/s, loss=0.0082,
Época 155/200 [Train] L1=8.91e-05: 100%| | 185/185 [00:02<00:00, 79.64it/s, loss=0.0071,

Época 155/200 | L1 coef: 8.91e-05
Train - Total: 0.0078 | MSE: 0.0029 | L0: 535.6
Val - Total: 0.0082 | MSE: 0.0032 | L0: 543.4
Best Val MSE: 0.0030 @ época 150 | Sin mejora: 5/20

Época 156/200 [Train] L1=8.73e-05: 100%| | 185/185 [00:02<00:00, 79.46it/s, loss=0.0069,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 156 (Val MSE: 0.0026)

Época 157/200 [Train] L1=8.56e-05: 100%| | 185/185 [00:02<00:00, 81.58it/s, loss=0.0065,
Época 158/200 [Train] L1=8.39e-05: 100%| | 185/185 [00:02<00:00, 82.83it/s, loss=0.0077,
Época 159/200 [Train] L1=8.22e-05: 100%| | 185/185 [00:02<00:00, 81.32it/s, loss=0.0073,
Época 160/200 [Train] L1=8.05e-05: 100%| | 185/185 [00:02<00:00, 81.03it/s, loss=0.0065,

Época 160/200 | L1 coef: 8.05e-05

Train - Total: 0.0073 | MSE: 0.0028 | L0: 536.6

Val - Total: 0.0076 | MSE: 0.0030 | L0: 543.7

Best Val MSE: 0.0026 @ época 156 | Sin mejora: 4/20

Época 161/200 [Train] L1=7.89e-05: 100%| | 185/185 [00:02<00:00, 78.52it/s, loss=0.0066,
Época 162/200 [Train] L1=7.73e-05: 100%| | 185/185 [00:02<00:00, 80.22it/s, loss=0.0064,
Época 163/200 [Train] L1=7.58e-05: 100%| | 185/185 [00:02<00:00, 78.68it/s, loss=0.0060,
Época 164/200 [Train] L1=7.43e-05: 100%| | 185/185 [00:02<00:00, 81.63it/s, loss=0.0062,
Época 165/200 [Train] L1=7.28e-05: 100%| | 185/185 [00:02<00:00, 80.38it/s, loss=0.0062,

Época 165/200 | L1 coef: 7.28e-05

Train - Total: 0.0061 | MSE: 0.0020 | L0: 537.7

Val - Total: 0.0068 | MSE: 0.0027 | L0: 544.4

Best Val MSE: 0.0026 @ época 156 | Sin mejora: 9/20

Época 166/200 [Train] L1=7.13e-05: 100%| | 185/185 [00:02<00:00, 79.82it/s, loss=0.0063,
Época 167/200 [Train] L1=6.99e-05: 100%| | 185/185 [00:02<00:00, 81.66it/s, loss=0.0062,
Época 168/200 [Train] L1=6.85e-05: 100%| | 185/185 [00:02<00:00, 80.93it/s, loss=0.0059,
Época 169/200 [Train] L1=6.71e-05: 100%| | 185/185 [00:02<00:00, 80.17it/s, loss=0.0059,
Época 170/200 [Train] L1=6.58e-05: 100%| | 185/185 [00:02<00:00, 79.77it/s, loss=0.0056,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 170 (Val MSE: 0.0026)

Época 170/200 | L1 coef: 6.58e-05

Train - Total: 0.0056 | MSE: 0.0018 | L0: 538.4

Val - Total: 0.0064 | MSE: 0.0026 | L0: 545.5

Best Val MSE: 0.0026 @ época 170 | Sin mejora: 0/20

Época 171/200 [Train] L1=6.45e-05: 100%| | 185/185 [00:02<00:00, 77.88it/s, loss=0.0060,
Época 172/200 [Train] L1=6.32e-05: 100%| | 185/185 [00:02<00:00, 78.10it/s, loss=0.0055,
Época 173/200 [Train] L1=6.19e-05: 100%| | 185/185 [00:02<00:00, 77.83it/s, loss=0.0055,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 173 (Val MSE: 0.0024)

Época 174/200 [Train] L1=6.07e-05: 100%| | 185/185 [00:02<00:00, 77.94it/s, loss=0.0058,

Época 175/200 [Train] L1=5.95e-05: 100%| | 185/185 [00:02<00:00, 80.46it/s, loss=0.0055,

Época 175/200 | L1 coef: 5.95e-05

Train - Total: 0.0059 | MSE: 0.0025 | L0: 538.6

Val - Total: 0.0063 | MSE: 0.0028 | L0: 546.2

Best Val MSE: 0.0024 @ época 173 | Sin mejora: 2/20

Época 176/200 [Train] L1=5.83e-05: 100%| | 185/185 [00:02<00:00, 78.72it/s, loss=0.0051,

Época 177/200 [Train] L1=5.71e-05: 100%| | 185/185 [00:02<00:00, 78.63it/s, loss=0.0054,

Época 178/200 [Train] L1=5.60e-05: 100%| | 185/185 [00:02<00:00, 80.77it/s, loss=0.0051,

Época 179/200 [Train] L1=5.49e-05: 100%| | 185/185 [00:02<00:00, 80.03it/s, loss=0.0050,

Época 180/200 [Train] L1=5.38e-05: 100%| | 185/185 [00:02<00:00, 79.62it/s, loss=0.0048,

Época 180/200 | L1 coef: 5.38e-05

Train - Total: 0.0053 | MSE: 0.0021 | L0: 539.1

Val - Total: 0.0057 | MSE: 0.0025 | L0: 546.3

Best Val MSE: 0.0024 @ época 173 | Sin mejora: 7/20

Época 181/200 [Train] L1=5.27e-05: 100%| | 185/185 [00:02<00:00, 77.83it/s, loss=0.0049,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 181 (Val MSE: 0.0024)

Época 182/200 [Train] L1=5.16e-05: 100%| | 185/185 [00:02<00:00, 79.54it/s, loss=0.0052,

Época 183/200 [Train] L1=5.06e-05: 100%| | 185/185 [00:02<00:00, 81.97it/s, loss=0.0051,

Época 184/200 [Train] L1=4.96e-05: 100%| | 185/185 [00:02<00:00, 81.35it/s, loss=0.0044,

Época 185/200 [Train] L1=4.86e-05: 100%| | 185/185 [00:02<00:00, 79.35it/s, loss=0.0042,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt

Mejor modelo guardado en época 185 (Val MSE: 0.0021)

Época 185/200 | L1 coef: 4.86e-05
Train - Total: 0.0044 | MSE: 0.0015 | L0: 540.0
Val - Total: 0.0050 | MSE: 0.0021 | L0: 546.6
Best Val MSE: 0.0021 @ época 185 | Sin mejora: 0/20

Época 186/200 [Train] L1=4.76e-05: 100%	185/185 [00:02<00:00, 78.01it/s, loss=0.0048,
Época 187/200 [Train] L1=4.67e-05: 100%	185/185 [00:02<00:00, 78.47it/s, loss=0.0048,
Época 188/200 [Train] L1=4.57e-05: 100%	185/185 [00:02<00:00, 76.27it/s, loss=0.0049,
Época 189/200 [Train] L1=4.48e-05: 100%	185/185 [00:02<00:00, 74.85it/s, loss=0.0045,
Época 190/200 [Train] L1=4.39e-05: 100%	185/185 [00:02<00:00, 78.99it/s, loss=0.0042,

Época 190/200 | L1 coef: 4.39e-05
Train - Total: 0.0044 | MSE: 0.0017 | L0: 540.3
Val - Total: 0.0051 | MSE: 0.0024 | L0: 547.2
Best Val MSE: 0.0021 @ época 185 | Sin mejora: 5/20

Época 191/200 [Train] L1=4.31e-05: 100%	185/185 [00:02<00:00, 79.08it/s, loss=0.0044,
Época 192/200 [Train] L1=4.22e-05: 100%	185/185 [00:02<00:00, 79.62it/s, loss=0.0044,
Época 193/200 [Train] L1=4.13e-05: 100%	185/185 [00:02<00:00, 76.20it/s, loss=0.0043,
Época 194/200 [Train] L1=4.05e-05: 100%	185/185 [00:02<00:00, 78.44it/s, loss=0.0039,
Época 195/200 [Train] L1=3.97e-05: 100%	185/185 [00:02<00:00, 79.38it/s, loss=0.0038,

Época 195/200 | L1 coef: 3.97e-05
Train - Total: 0.0039 | MSE: 0.0015 | L0: 540.6
Val - Total: 0.0046 | MSE: 0.0021 | L0: 548.0
Best Val MSE: 0.0021 @ época 185 | Sin mejora: 10/20

Época 196/200 [Train] L1=3.89e-05: 100%	185/185 [00:02<00:00, 79.23it/s, loss=0.0042,
Época 197/200 [Train] L1=3.81e-05: 100%	185/185 [00:02<00:00, 77.47it/s, loss=0.0039,
Época 198/200 [Train] L1=3.74e-05: 100%	185/185 [00:02<00:00, 76.38it/s, loss=0.0041,
Época 199/200 [Train] L1=3.66e-05: 100%	185/185 [00:02<00:00, 77.69it/s, loss=0.0041,
Época 200/200 [Train] L1=3.59e-05: 100%	185/185 [00:02<00:00, 78.98it/s, loss=0.0040,

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_best.pt
Mejor modelo guardado en época 200 (Val MSE: 0.0021)

Época 200/200 | L1 coef: 3.59e-05
Train - Total: 0.0037 | MSE: 0.0015 | L0: 541.0

Val - Total: 0.0043 | MSE: 0.0021 | L0: 547.9
Best Val MSE: 0.0021 @ época 200 | Sin mejora: 0/20

=====

RESUMEN DEL ENTRENAMIENTO

=====

Mejor modelo en época 200 con Val MSE = 0.0021
Early stopping tras 200 épocas

Métricas Finales (Última Época):

Métrica	Train	Validation
Total Loss	0.0037	0.0043
MSE Loss	0.0015	0.0021
L0 Sparsity	541.0	547.9
Fraction Active	3.30%	3.34%

=====

8. Guardar Modelo

```
# Guardar checkpoint final (local + organizado)
checkpoint = {
    'model_state_dict': model.state_dict(),
    'optimizer_state_dict': optimizer.state_dict(),
    'config': CONFIG,
    'history': history
}

# Guardado local
final_model_name = get_model_name(
    variante=NAMING_META['variante'],
    tecnica=NAMING_META['tecnica'],
    capa=NAMING_META['capa'],
    juegos=NAMING_META['juegos'],
    estado='final'
)
checkpoint_path = CONFIG['save_dir'] / final_model_name
torch.save(checkpoint, checkpoint_path)

# Guardado organizado
save_checkpoint(
    model,
```

```

base_path=NAMING_META['base_path'],
variante=NAMING_META['variante'],
tecnica=NAMING_META['tecnica'],
capa=NAMING_META['capa'],
juegos=NAMING_META['juegos'],
estado='final',
extras=NAMING_META['extras'],
config=CONFIG,
history=history,
optimizer=optimizer
)

# Resumen de archivos guardados
saved_files = pd.DataFrame({
    'Tipo': ['Local', 'Organizado'],
    'Ubicación': [
        str(checkpoint_path),
        f"layer_{NAMING_META['capa']:02d}/models/..."
    ]
})

print('\n' + '='*70)
print(' CHECKPOINTS GUARDADOS')
print('='*70)
print(saved_files.to_string(index=False))
print('='*70)

```

Checkpoint guardado: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_standard\1000games\sae_standard_l1-decay_l6_1000g_ex3_final.pt

```

=====
CHECKPOINTS GUARDADOS
=====

```

Tipo	Ubicación
Local	saved_model\sae_standard_l1-decay_l6_1000g_final.pt
Organizado	layer_06/models/...

```

=====

```

9. Métricas básicas del modelo

```
# Guardar metricas de entrenamiento
metrics_dir = get_metrics_dir(
    NAMING_META["experiment_base_path"],
    NAMING_META["variante"],
    NAMING_META["tecnica"],
    NAMING_META["capa"],
    NAMING_META["juegos"],
    extras_id=extras_id
)

training_metrics = {
    # Metricas de calidad
    "val_mse_final": history["val_mse_loss"][-1],
    "val_mse_best": best_val_mse,
    "val_l0_final": history["val_l0_sparsity"][-1],
    "val_fraction_active_final": history["val_fraction_active"][-1],
    # Contexto del entrenamiento
    "best_epoch": best_epoch,
    "total_epochs": len(history["train_total_loss"]),
    "train_mse_final": history["train_mse_loss"][-1],
    "train_l0_final": history["train_l0_sparsity"][-1],
    # Hiperparametros
    "hidden_dim": CONFIG["hidden_dim"],
    "expansion_factor": CONFIG["hidden_dim"] / CONFIG["input_dim"],
    "sparsity_coef": CONFIG["sparsity_coef"],
    "sparsity_decay": CONFIG["sparsity_decay"],
    "learning_rate": CONFIG["learning_rate"],
    "num_games": NAMING_META["juegos"],
}

metrics_path = metrics_dir / "training_metrics.json"
with open(metrics_path, "w") as f:
    _json.dump(training_metrics, f, indent=2)

# Resumen
metrics_df = pd.DataFrame({
    "Metrica": ["Val MSE (final)", "Val MSE (mejor)", "Val L0 (final)", "Fraccion Activa"],
    "Valor": [
        f"{training_metrics['val_mse_final']:.6f}",
        f"{training_metrics['val_mse_best']:.6f}",

```

```

        f"{training_metrics['val_l0_final']:.1f}",
        f"{training_metrics['val_fraction_active_final']:.2%}"
    ]
})
print(" " + "="*50)
print(" METRICAS DE ENTRENAMIENTO GUARDADAS")
print("="*50)
print(metrics_df.to_string(index=False))
print(f"Archivo: {metrics_path}")
print("="*50)

```

```

=====
METRICAS DE ENTRENAMIENTO GUARDADAS
=====

```

	Metrica	Valor
Val MSE (final)	0.002122	
Val MSE (mejor)	0.002122	
Val L0 (final)	547.9	
Fraccion Activa	3.34%	
Archivo:	c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\1000games\ex3\metrics\training_metrics.json	

```

=====

```

10. Visualización de Resultados

```

# Gráficas de pérdidas (Train vs Validation) + L1 Decay
fig, axes = plt.subplots(2, 3, figsize=(18, 10))

# Total loss
axes[0, 0].plot(history['train_total_loss'], label='Train', alpha=0.8)
axes[0, 0].plot(history['val_total_loss'], label='Validation', alpha=0.8)
axes[0, 0].set_title('Total Loss')
axes[0, 0].set_xlabel('Época')
axes[0, 0].set_ylabel('Loss')
axes[0, 0].legend()
axes[0, 0].grid(True, alpha=0.3)

# MSE loss
axes[0, 1].plot(history['train_mse_loss'], label='Train', color='orange', alpha=0.8)
axes[0, 1].plot(history['val_mse_loss'], label='Validation', color='red', alpha=0.8)

```

```

if CONFIG['early_stopping'] and best_epoch > 0:
    axes[0, 1].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Best')
axes[0, 1].set_title('MSE Loss (Reconstrucción)')
axes[0, 1].set_xlabel('Época')
axes[0, 1].set_ylabel('MSE')
axes[0, 1].legend()
axes[0, 1].grid(True, alpha=0.3)

# L0 sparsity
axes[0, 2].plot(history['train_l0_sparsity'], label='Train', color='green', alpha=0.8)
axes[0, 2].plot(history['val_l0_sparsity'], label='Validation', color='darkgreen', alpha=0.8)
axes[0, 2].set_title('L0 Sparsity (Features Activas)')
axes[0, 2].set_xlabel('Época')
axes[0, 2].set_ylabel('Número de features')
axes[0, 2].legend()
axes[0, 2].grid(True, alpha=0.3)

# Fraction active
axes[1, 0].plot(history['train_fraction_active'], label='Train', color='purple', alpha=0.8)
axes[1, 0].plot(history['val_fraction_active'], label='Validation', color='magenta', alpha=0.8)
axes[1, 0].set_title('Fracción de Features Activas')
axes[1, 0].set_xlabel('Época')
axes[1, 0].set_ylabel('Fracción')
axes[1, 0].legend()
axes[1, 0].grid(True, alpha=0.3)

# Sparsity Coefficient (L1 Decay)
axes[1, 1].plot(history['sparsity_coef_history'], color='brown', linewidth=2)
axes[1, 1].set_title('L1 Coefficient (Decay)')
axes[1, 1].set_xlabel('Época')
axes[1, 1].set_ylabel('Sparsity Coef')
axes[1, 1].set_yscale('log')
axes[1, 1].grid(True, alpha=0.3)

# Train vs Val MSE Gap
mse_gap = [v - t for v, t in zip(history['val_mse_loss'], history['train_mse_loss'])]
axes[1, 2].plot(mse_gap, color='crimson', linewidth=2)
axes[1, 2].axhline(0, color='black', linestyle='--', alpha=0.5)
if CONFIG['early_stopping'] and best_epoch > 0:
    axes[1, 2].axvline(best_epoch - 1, color='green', linestyle='--', alpha=0.7, label=f'Best')
axes[1, 2].set_title('Overfitting Gap (Val MSE - Train MSE)')
axes[1, 2].set_xlabel('Época')

```

```

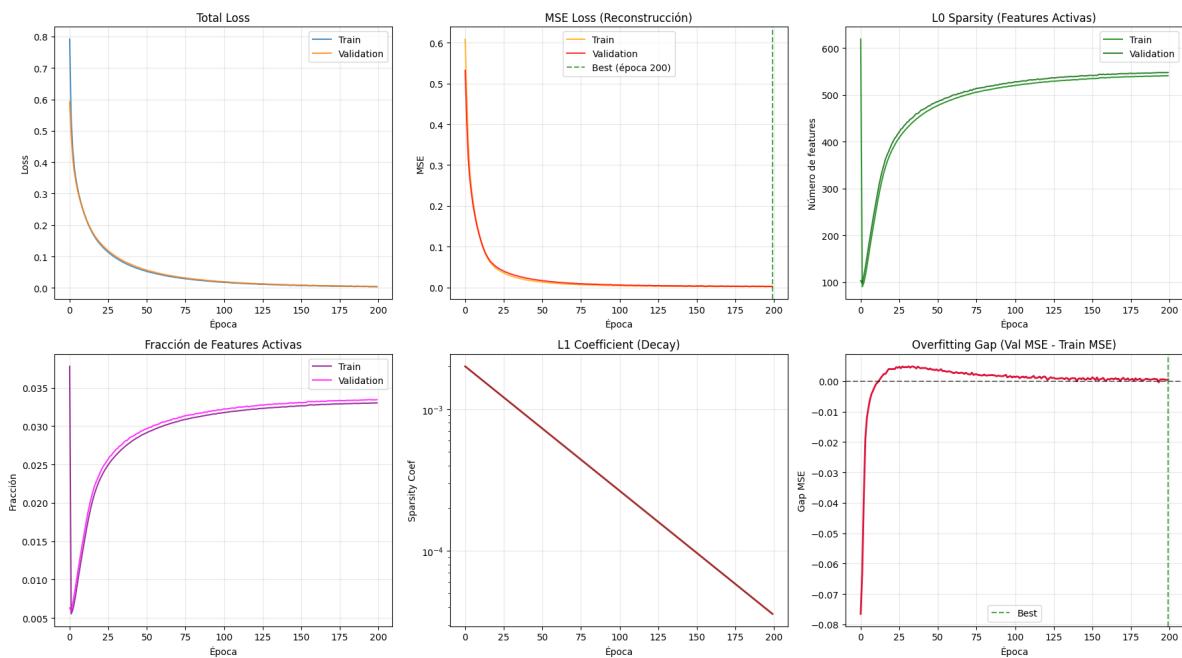
axes[1, 2].set_ylabel('Gap MSE')
axes[1, 2].legend()
axes[1, 2].grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig(CONFIG['plots_dir'] / 'training_curves.png', dpi=300, bbox_inches='tight')
plt.show()

print('\n' + '='*50)
print('  Gráficas guardadas exitosamente')

print('='*50)

```



=====

Gráficas guardadas exitosamente

=====

11. Resumen Final


```

# Resumen final completo
print('\n' + '='*70)
print(' RESUMEN FINAL DEL ENTRENAMIENTO')
print('='*70)

# Arquitectura
arch_summary = pd.DataFrame({
    'Componente': ['Input', 'Hidden', 'Parámetros'],
    'Valor': [
        f"{CONFIG['input_dim']}",
        f"{CONFIG['hidden_dim']} ({CONFIG['hidden_dim']/CONFIG['input_dim']:.1f}x)",
        f"{num_params:}"
    ]
})
print('\nArquitectura:')
print(arch_summary.to_string(index=False))

# Datos
data_summary = pd.DataFrame({
    'Conjunto': ['Train', 'Validation', 'Total'],
    'Muestras': [
        f"{len(train_dataset):}",
        f"{len(val_dataset):}",
        f"{len(full_dataset):}"
    ]
})
print('\nDatos:')
print(data_summary.to_string(index=False))

# Métricas finales
final_summary = pd.DataFrame({
    'Métrica': ['Train Loss', 'Val Loss', 'Train MSE', 'Val MSE', 'Train L0', 'Val L0'],
    'Valor': [
        f"{history['train_total_loss'][-1]:.4f}",
        f"{history['val_total_loss'][-1]:.4f}",
        f"{history['train_mse_loss'][-1]:.4f}",
        f"{history['val_mse_loss'][-1]:.4f}",
        f"{history['train_l0_sparsity'][-1]:.1f}",
        f"{history['val_l0_sparsity'][-1]:.1f}"
    ]
})
print('\nMétricas Finales:')

```

```

print(final_summary.to_string(index=False))

# Archivos
files_summary = pd.DataFrame({
    'Tipo': ['Modelo', 'Gráficas'],
    'Ubicación': [
        str(checkpoint_path),
        str(CONFIG['plots_dir'] / 'training_curves.png')
    ]
})
print('\nArchivos Guardados:')
print(files_summary.to_string(index=False))

print('\n' + '='*70)
print('  ENTRENAMIENTO COMPLETADO EXITOSAMENTE')
print('='*70)

```

```

=====
RESUMEN FINAL DEL ENTRENAMIENTO
=====

```

Arquitectura:

Componente	Valor
Input	512
Hidden	16384 (32.0x)
Parámetros	16,794,112

Datos:

Conjunto	Muestras
Train	47,200
Validation	11,800
Total	59,000

Métricas Finales:

Métrica	Valor
Train Loss	0.0037
Val Loss	0.0043
Train MSE	0.0015
Val MSE	0.0021
Train L0	541.0
Val L0	547.9

Archivos Guardados:

Tipo

Modelo

decay_l6_1000g_final.pt

Gráficas c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard

decay_standard\1000games\ex3\plots\training_curves.png

```
=====
ENTRENAMIENTO COMPLETADO EXITOSAMENTE
=====
```

12. Generar PDF con Quarto

Una vez ejecutado todo el notebook, ejecutar el siguiente comando en la terminal para generar el PDF con Quarto:

```
import subprocess

# Obtener directorio del experimento y nombre del PDF
exp_dir = get_experiment_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'training',
    extras_id=extras_id
)

notebook_name = 'train_sae_standard_l1-decay.ipynb'
output_path = exp_dir / pdf_name

print(f' Generando PDF con Quarto...')
```

```

print(f' Archivo de salida: {output_path}')

# Quarto no acepta rutas en --output, solo nombre de archivo
# Se usa --output-dir para el directorio y --output solo para el nombre
result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{exp_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
else:
    print(f' Error al generar PDF:')
    print(result.stderr)

```

Generando PDF con Quarto...

Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\1000games\ex3\sae_standard_l1-decay_l6_1000g_ex3_training.pdf

PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_standard\1000games\ex3\sae_standard_l1-decay_l6_1000g_ex3_training.pdf